

Krishna Subedi

Applied AI Engineer — LLM Systems & Robotics

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EXPERIENCE

Neryva

Jul 2025 – Present

Member of Technical Staff · Remote

- Architected the production multi-tenant agent runtime: asynchronous orchestration of agent lifecycles, tool-calling loops, and state-machine transitions with hierarchical memory — LLM output treated as untrusted code, executed in isolated sandboxes to run many agents concurrently per customer.
- Built agent harnesses enforcing input/output safety (classifier-based ingress checks for prompt injection and jailbreaks, egress validation) and evaluation pipelines for tool-calling accuracy and retrieval-context drift. Demo: neryva.com.

Smaitic Labs

Apr 2024 – Jul 2025

Full-stack Developer

- Built a three-tier admin hierarchy (main/state/seller) with role-specific access and per-tier dashboards, with a multi-tenant Spring Boot backend (hierarchical RBAC, ZipCode-based delivery logistics) that accommodates new regions and sellers without structural change and runs an automated commission engine that eliminates manual per-order fee calculation.
- Shipped [Vanidya](#) marketplace applications across native Android and an offline-first PWA, featuring real-time order tracking.

Nepal Oil Corporation Limited

Oct 2023 – Mar 2024

Software Engineer

- Designed a time-series forecasting system for regional fuel demand, modeling surges, strikes, and protests so inventory allocation tracked real consumption spikes, supply stayed smooth, and swings never reached the pump.
- Built an automated dispatch reporting system for fuel logistics, consolidating daily depot and tanker movements into live dashboards that replaced manual end-of-day report preparation.

PUBLICATIONS

Bounded Precision-Geometry Scaling for Robust Multi-Task Learning

Aug 2026

Sole Author · [arXiv:2608.21653](https://arxiv.org/abs/2608.21653)

- Held macro $0.777 \rightarrow 0.778$ under $1000\times$ loss rescaling, outperforming Nash-MTL on NYUv2.

The Reliability of LLMs for Medical Diagnosis

Mar 2025

Sole Author · [arXiv:2503.10647](https://arxiv.org/abs/2503.10647)

- Evaluated leading LLMs on 52 patient cases; quantified sensitivity to misleading prompts and contextual manipulations.

PROJECTS

Agent Studio: Enterprise AI Agent Platform

Jul 2025 – Present

Work Project & Open Source · ML & Platform

- Architected a multi-tenant LLM control plane based on the CoALA framework, integrating scoped memory, action spaces, and a LangGraph decision loop, alongside vector-isolated RAG and an LLM gateway for provider routing, fallback chains, circuit breakers, and granular cost tracking.
- Built the session/context engine (durable threads, compaction, conversation forking) with Presidio PII redaction, NeMo Guardrails, and human-in-the-loop escalation, plus an evaluation stack using Garak scans and RAGAS suites.

Krishi Vaidya: AI-Powered Crop Disease Diagnosis

Apr – Jul 2026

Personal Project · ML

- Engineered a dual-model crop-disease diagnosis system combining an edge-deployed YOLO detector for real-time symptom localization with a cloud-deployed Qwen2.5-VL-3B VLM, fine-tuned via LoRA/PEFT, for structured multimodal diagnosis.
- Optimized multi-tier inference by benchmarking the YOLO detector across INT4, INT8, and FP8 configurations, deploying an INT8 TFLite model for on-device OpenCV inference, and serving the VLM via vLLM with 4-bit NF4 quantization and FlashAttention-2, enforcing structured JSON outputs with confidence scoring and OOD detection.

EDUCATION

Bachelor of Engineering in Mechanical Engineering

2019 – 2023

Tribhuvan University · Licensed PE

SKILLS

Languages: Python, TypeScript/JavaScript, Java, Kotlin, C/C++, SQL · **AI/ML:** PyTorch, Transformers, LLMs/VLMs, LoRA/PEFT, Computer Vision, Time-Series, Multi-Task & Continual Learning · **LLM Systems:** RAG, LangGraph, LLM/Agent Evaluation, Tool Calling, vLLM, Guardrails · **Full-Stack & Web:** React, Spring Boot, FastAPI, Node.js (Express), Redux · **Infrastructure:** Docker, Kubernetes, Redis, PostgreSQL, Git, CI/CD, Linux